Differentiation & Integration

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

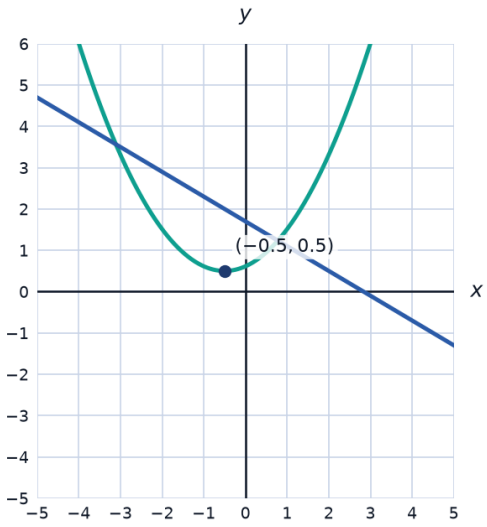
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Differentiation, Rules & Rates of Change · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Integration, Areas & Kinematics · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

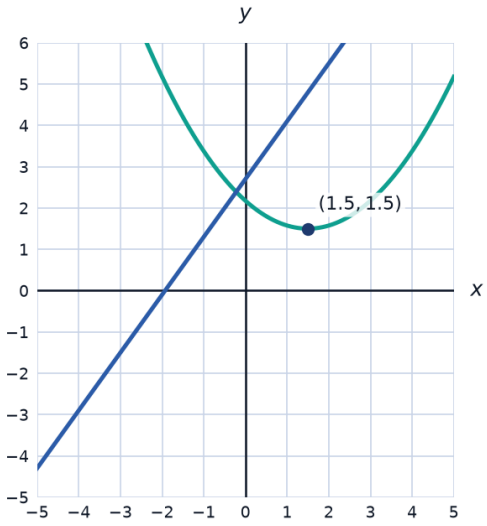
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of t and the value at $t = 1$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

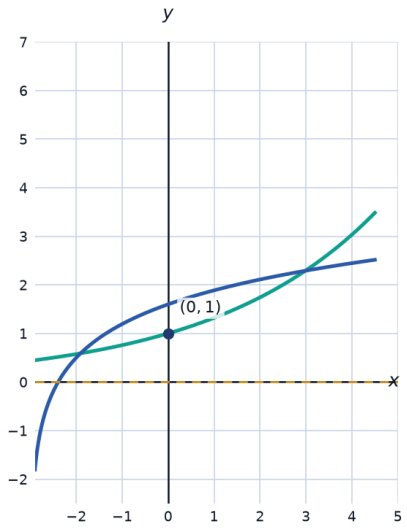
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

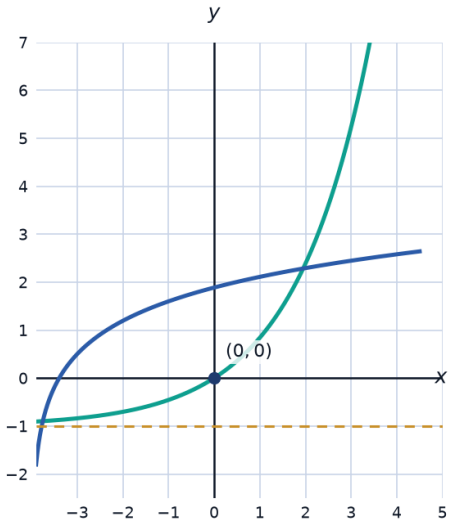
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Graphs & Transformations**

[Logarithms, Exponentials & Straight-Line Reduction · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Coordinate Geometry**

[AI SL — Specific Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry**

[Overview · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability**

[Number & Algebra · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Functions: Composite & Inverse**

[Calculus · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. 8

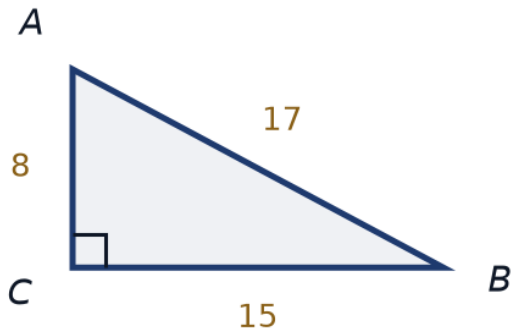
b. 12

c. 4

d. 0

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.



a. $y = 3x - 7$

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. Local max at $(1, 6)$, local min at $(2, 5)$

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ or $x > 3$

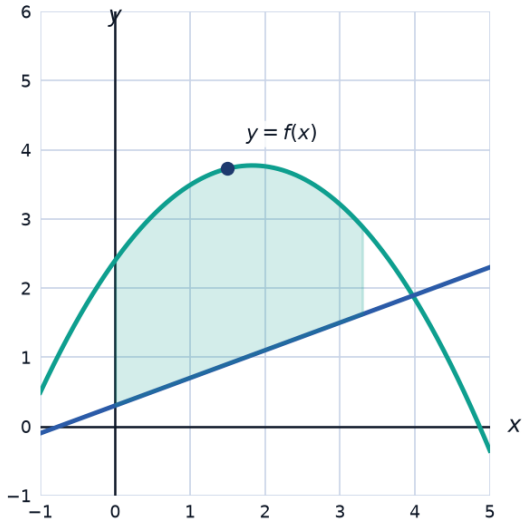
b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.



a. 18

b. 24

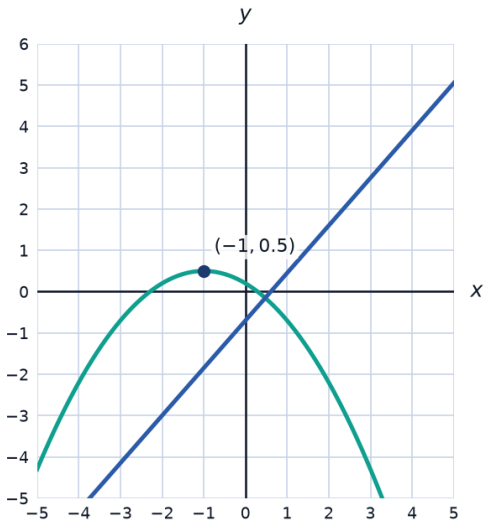
c. 16

d. 20

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.



a. $y = 2x^2 - x + 3$

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. 0

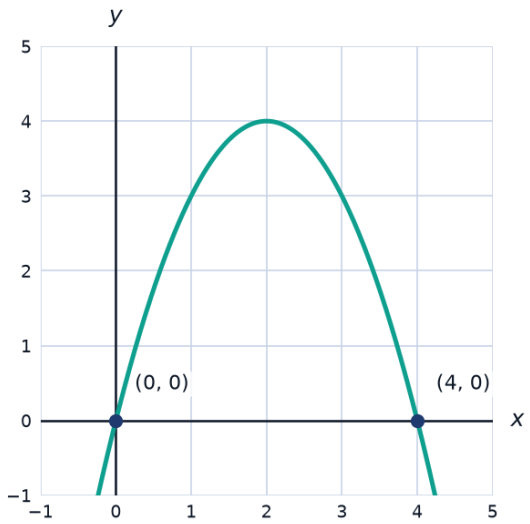
b. 9

c. 18

d. 4.5

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



a. $\frac{32}{3}$

b. 16

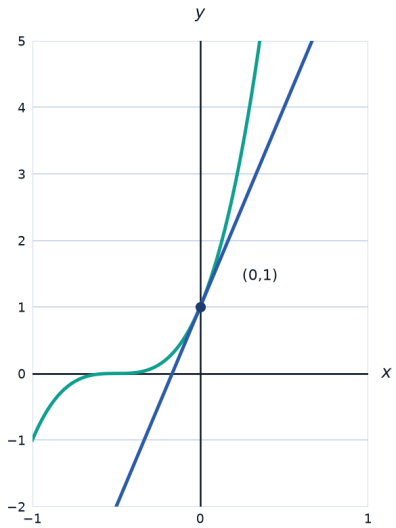
c. $\frac{64}{3}$

d. $\frac{16}{3}$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.



a. $y = 2x + 1$

b. $y = 6x - 1$

c. $y = 6x + 1$

d. $y = 6x + 3$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.

a. $(1, 6)$, which is a local minimum

b. $(1, 6)$, which is a local maximum

c. $(-1, -2)$, which is a local minimum

d. $(2, 10)$, which is a local minimum

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm.

a. 20π

b. 50π

c. $\frac{36\pi}{5}$

d. 11π

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The variables x and y are connected by the equation $y = \sqrt{2x + 5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

a. 2 units per second

b. 1 unit per second

c. $\frac{1}{3}$ units per second

d. 12 units per second

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

a. 2

b. 4

c. 3

d. 1

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Circular Measure

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$

b. $6e^3$

c. $3e^3$

d. $2e^3$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

a. 57 units per second

b. 19 units per second

c. $\frac{19}{3}$ units per second

d. 21 units per second

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Series & Binomial Theorem

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.

a. 76 units s^{-1}

b. 19 units s^{-1}

c. 28 units s^{-1}

d. 24 units s^{-1}

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Kinematics

A curve has equation $y = \frac{kx}{2x - 1}$, where $x \neq \frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

a. -0.2 units per second

b. 0.2 units per second

c. -1.8 units per second

d. -0.6 units per second

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Forces & Equilibrium

Find the exact value of the definite integral $\int_0^{\ln 2} \text{e}^{2x}\,\text{d}x$.

a. $\frac{3}{2}$

b. 3

c. $\frac{1}{2}$

d. $\frac{7}{2}$

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